



**Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
*Institut Teknologi Sepuluh Nopember***

Laporan Akhir Praktikum Jaringan Komputer

Vlan–Trunk–OSPF–MultiVendor

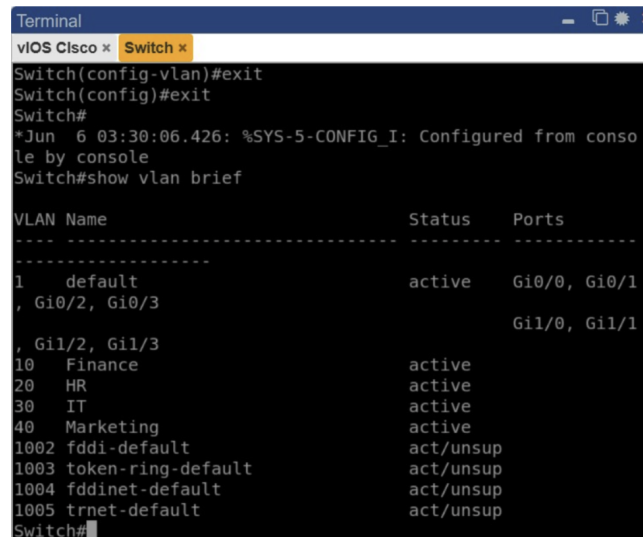
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2026

1 Langkah-Langkah Percobaan

1.1 Konfigurasi Cisco Switch

1. Masuk ke terminal Cisco Switch
2. Buat VLAN 10, 20, 30, dan 40, lalu verifikasi apakah statusnya sudah aktif

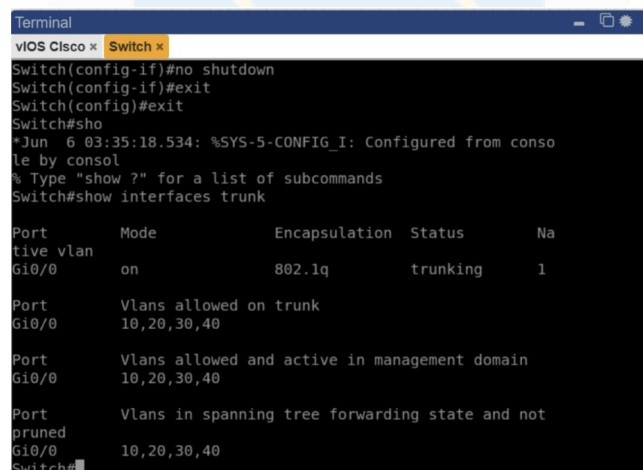


```
Terminal
vIOS Cisco x Switch x
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
*Jun 6 03:30:06.426: %SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Gi0/0, Gi0/1, Gi0/2, Gi0/3, Gi1/0, Gi1/1, Gi1/2, Gi1/3
10   Finance                 active
20   HR                      active
30   IT                      active
40   Marketing               active
1002 fddi-default           act/unsup
1003 token-ring-default     act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default         act/unsup
Switch#
```

Gambar 1: Verifikasi VLAN

3. Konfigurasi *access port* untuk setiap VLAN sesuai dengan *interface* pada Cisco Switch
4. Konfigurasi *trunk* dari switch menuju Cisco Router untuk VLAN 10, 20, 30, dan 40, lalu verifikasi dan *save* semua perintah



```
Terminal
vIOS Cisco x Switch x
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#exit
Switch#sho
*Jun 6 03:35:18.534: %SYS-5-CONFIG_I: Configured from console by console
% Type "show ?" for a list of subcommands
Switch#show interfaces trunk

Port        Mode          Encapsulation  Status        Native vlan
Gi0/0       on            802.1q         trunking      1

Port        Vlans allowed on trunk
Gi0/0       10,20,30,40

Port        Vlans allowed and active in management domain
Gi0/0       10,20,30,40

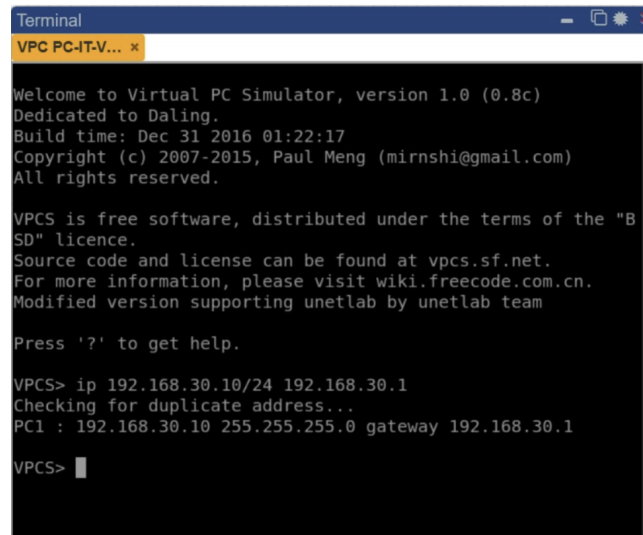
Port        Vlans in spanning tree forwarding state and not pruned
Gi0/0       10,20,30,40
Switch#
```

Gambar 2: Verifikasi Trunk Port

1.2 Konfigurasi Cisco Router

1. Masuk ke Cisco Router
2. Konfigurasi *interface* trunk menuju Cisco Switch

3. Konfigurasi *subinterface* untuk VLAN 10, 20, 30, dan 40 menggunakan metode *Router on a Stick*
4. Konfigurasi DHCP Server untuk VLAN 10 dan VLAN 20
5. Konfigurasi IP *static* pada VPCS untuk VLAN 30 dan VLAN 40, serta DHCP untuk VLAN 10 dan VLAN 20



```
Terminal
VPC PC-IT-V... x

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

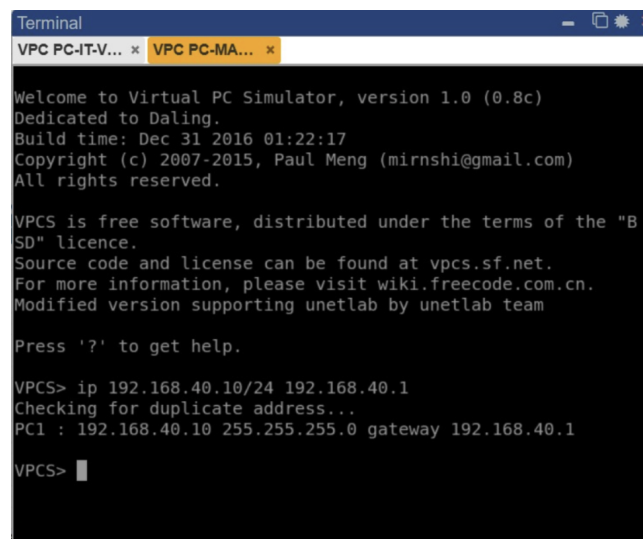
VPCS is free software, distributed under the terms of the "B
SD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS> 
```

Gambar 3: Konfigurasi IP Static pada PC VLAN 30 (IT)



```
Terminal
VPC PC-IT-V... x VPC PC-MA... x

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "B
SD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
PC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VPCS> 
```

Gambar 4: Konfigurasi IP static pada PC VLAN 40 (Marketing)

```

Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x VIOS Cisco x

Press '?' to get help.

VPCS> ip dhcp
DDORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS> ip dhcp
DDORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 86397, 86400/43200/75600
MAC        : 00:50:79:66:68:9b
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS>

```

Gambar 5: PC VLAN 10 mendapatkan IP via DHCP

```

Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x VIOS Cisco x

Press '?' to get help.

VPCS> ip dhcp
DDORA IP 192.168.20.11/24 GW 192.168.20.1

VPCS> ip dhcp
DDORA IP 192.168.20.11/24 GW 192.168.20.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.20.11/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE  : 86395, 86400/43200/75600
MAC        : 00:50:79:66:68:9a
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS>

```

Gambar 6: PC VLAN 20 mendapatkan IP via DHCP

6. Verifikasi *interface*, *DHCP pool*, dan *DHCP binding* pada Cisco Router

```

Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x VIOS Cisco x

*Jun  6 03:49:22.982: %SYS-5-CONFIG_I: Configured from console by console
% Type "show ?" for a list of subcommands
Router#show ip interface brief
Interface                                IP-Address      OK? Method Status
-----
GigabitEthernet0/0                      unassigned      YES unset  administratively down down
GigabitEthernet0/1                      unassigned      YES unset  administratively down down
GigabitEthernet0/2                      unassigned      YES unset  up
GigabitEthernet0/2.10                   192.168.10.1    YES manual  up
GigabitEthernet0/2.20                   192.168.20.1    YES manual  up
GigabitEthernet0/2.30                   192.168.30.1    YES manual  up
GigabitEthernet0/2.40                   192.168.40.1    YES manual  up
GigabitEthernet0/3                      unassigned      YES unset  administratively down down
Router#

```

Gambar 7: Verifikasi VLAN 10, 20, 30, 40

```

Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x

Pool VLAN10-FINANCE :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased ad
dresses
192.168.10.12 192.168.10.1 - 192.168.10.254 1

Pool VLAN20-HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased ad
dresses
192.168.20.12 192.168.20.1 - 192.168.20.254 1
Router#

```

Gambar 8: Verifikasi pool VLAN 10 dan VLAN 20

7. Lakukan pengujian dari setiap VPCS dengan *ping* ke *gateway* masing-masing dan *ping* antar-VLAN

```

Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x

VPCS> show ip
NAME       : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 86397, 86400/43200/75600
MAC        : 00:50:79:66:68:9b
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.10.1

64 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.135 ms
64 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=2.610 ms
64 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.629 ms
64 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=3.602 ms
64 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=3.202 ms

VPCS>

```

Gambar 9: PC VLAN 10 ping gateway 192.168.10.1

```

Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x

VPCS> show ip
NAME       : VPCS[1]
IP/MASK    : 192.168.20.11/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE  : 86395, 86400/43200/75600
MAC        : 00:50:79:66:68:9a
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.20.1

64 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=2.904 ms
64 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.744 ms
64 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.960 ms
64 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=3.726 ms
64 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=3.562 ms

VPCS>

```

Gambar 10: PC VLAN 20 ping gateway 192.168.20.1

```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x
DHCP LEASE : 86395, 86400/43200/75600
MAC : 00:50:79:66:68:9a
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500

VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=2.904 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.744 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.960 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=3.726 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=3.562 ms

VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=3.084 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=3.336 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=2.880 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=3.303 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=3.262 ms

VPCS>
```

Gambar 11: PC VLAN 20 ping gateway VLAN 20 dan VLAN 30

```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=2.904 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.744 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.960 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=3.726 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=3.562 ms

VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=3.084 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=3.336 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=2.880 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=3.303 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=3.262 ms

VPCS> ping 192.168.40.1

84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=3.172 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=3.483 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=3.394 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=3.611 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=4.493 ms

VPCS>
```

Gambar 12: PC VLAN 20 ping gateway VLAN 30 dan VLAN 40

```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x
DHCP LEASE : 86397, 86400/43200/75600
MAC : 00:50:79:66:68:9b
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500

VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.135 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=2.610 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.629 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=3.602 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=3.202 ms

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=10.095 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=5.725 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=4.266 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=5.099 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=5.338 ms

VPCS>
```

Gambar 13: PC VLAN 10 ping gateway dan ping ke VLAN 20

```

Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x VIOS Cisco x
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=10.095 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=5.725 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=4.266 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=5.099 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=5.338 ms

VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=11.953 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=5.356 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=6.404 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=5.350 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=4.446 ms

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=12.243 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=5.812 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=5.189 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=5.945 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=6.182 ms

VPCS>

```

Gambar 14: PC VLAN 10 ping ke VLAN 30 dan VLAN 40

```

Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x VIOS Cisco x
dresses
192.168.20.12 192.168.20.1 - 192.168.20.254 1
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address Client-ID/ Lease expiration
Type
Hardware address/
User name
192.168.10.11 0100.5079.6668.9b Jun 07 2026 03:48 AM
Automatic
192.168.20.11 0100.5079.6668.9a Jun 07 2026 03:48 AM
Automatic
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address Client-ID/ Lease expiration
Type
Hardware address/
User name
192.168.10.11 0100.5079.6668.9b Jun 07 2026 03:50 AM
Automatic
192.168.20.11 0100.5079.6668.9a Jun 07 2026 03:51 AM
Automatic
Router#

```

Gambar 15: Verifikasi dhcp binding

1.3 Konfigurasi Huawei Router dan IP Address Antarrouter

1. Masuk ke terminal Huawei Router
2. Konfigurasi LAN pada sisi Huawei dengan *network* 192.168.50.0/24

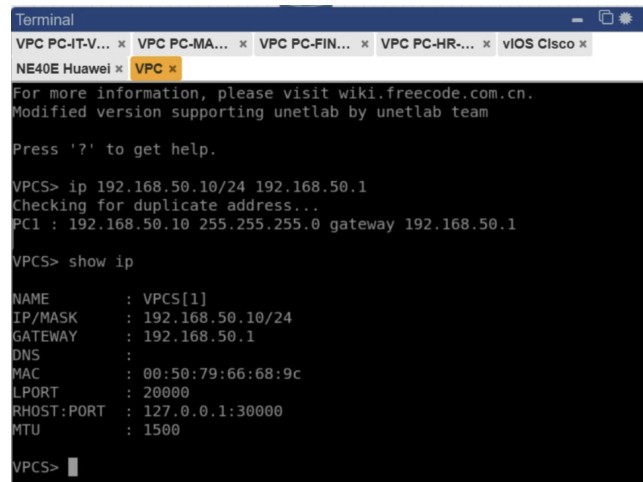
```

Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x VIOS Cisco x
NE40E Huawei x VPC x
<HUAWEI>system-view
Enter system view, return user view with return command.
[~HUAWEI]interface Ethernet1/0/2
[~HUAWEI-Ethernet1/0/2]description LAN-HUAWEI
[*HUAWEI-Ethernet1/0/2]ip address 192.168.50.1 255.255.255.0
[*HUAWEI-Ethernet1/0/2]undo shutdown
[*HUAWEI-Ethernet1/0/2]quit
[*HUAWEI]commit
[~HUAWEI]interface Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0]description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0]ip address 10.10.10.2
^
Error: Incomplete command found at '^' position.
[*HUAWEI-Ethernet1/0/0]ip address 10.10.10.2 255.255.255.252
[*HUAWEI-Ethernet1/0/0]undo shutdown
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]commit
[~HUAWEI]

```

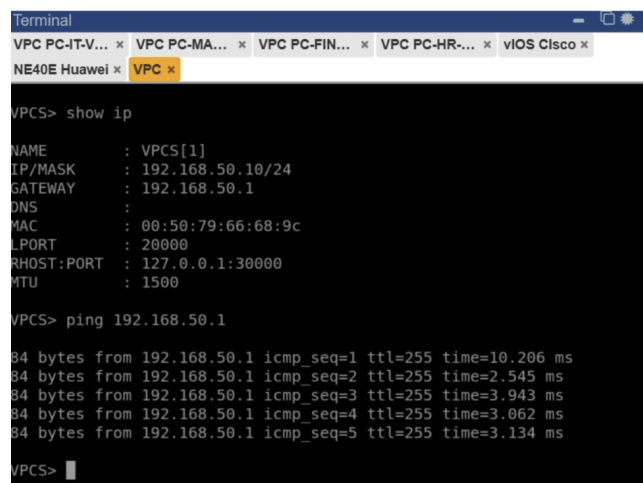
Gambar 16: Konfigurasi interface LAN Huawei

3. Konfigurasi IP *static* pada VPC LAN Huawei, lalu uji *ping* ke *gateway*



```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x
NE40E Huawei x VPC x
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team
Press '?' to get help.
VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1
VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 192.168.50.10/24
GATEWAY    : 192.168.50.1
DNS        :
MAC        : 00:50:79:66:68:9c
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500
VPCS>
```

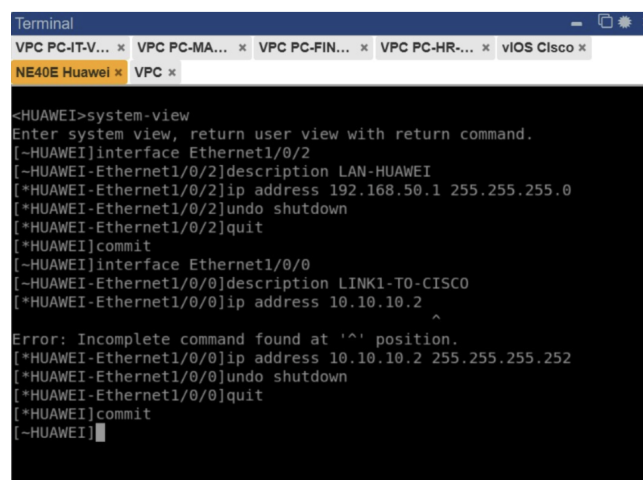
Gambar 17: Konfigurasi IP static pada PC LAN Huawei



```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x
NE40E Huawei x VPC x
VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 192.168.50.10/24
GATEWAY    : 192.168.50.1
DNS        :
MAC        : 00:50:79:66:68:9c
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500
VPCS> ping 192.168.50.1
64 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=10.206 ms
64 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=2.545 ms
64 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=3.943 ms
64 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=3.062 ms
64 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=3.134 ms
VPCS>
```

Gambar 18: PC LAN Huawei ping gateway

4. Konfigurasi alamat IP pada *link* Cisco ↔ Huawei *Link 1* dan *Link 2*

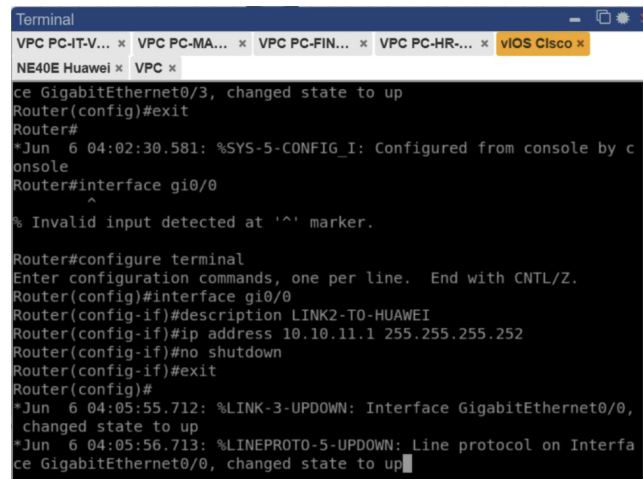


```
Terminal
VPC PC-IT-V... x VPC PC-MA... x VPC PC-FIN... x VPC PC-HR... x vIOS Cisco x
NE40E Huawei x VPC x
<HUAWEI>system-view
Enter system view, return user view with return command.
[~HUAWEI]interface Ethernet1/0/2
[~HUAWEI-Ethernet1/0/2]description LAN-HUAWEI
[*HUAWEI-Ethernet1/0/2]ip address 192.168.50.1 255.255.255.0
[*HUAWEI-Ethernet1/0/2]undo shutdown
[*HUAWEI-Ethernet1/0/2]quit
[*HUAWEI]commit
[~HUAWEI]interface Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0]description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0]ip address 10.10.10.2
Error: Incomplete command found at '^' position.
[*HUAWEI-Ethernet1/0/0]ip address 10.10.10.2 255.255.255.252
[*HUAWEI-Ethernet1/0/0]undo shutdown
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]commit
[~HUAWEI]
```

Gambar 19: Konfigurasi Ethernet1/0/0 Huawei

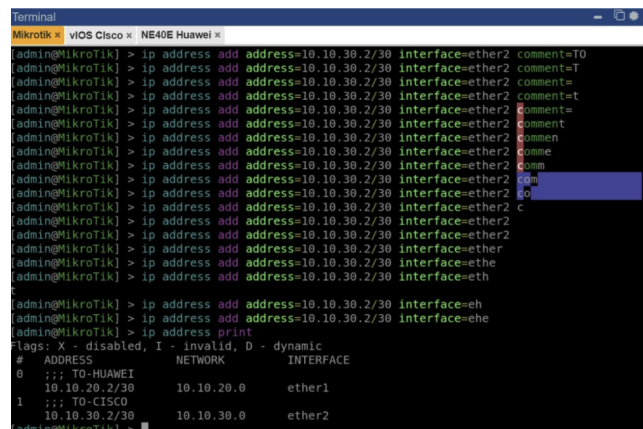
5. Konfigurasi alamat IP pada *link* Huawei ↔ MikroTik

6. Konfigurasi alamat IP pada *link* Cisco ↔ MikroTik



```
Router#
Router(config)#interface gi0/0
Router(config-if)#description LINK2-T0-HUAWEI
Router(config-if)#ip address 10.10.11.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 6 04:02:30.581: %SYS-5-CONFIG_I: Configured from console by c
onsole
Router#interface gi0/0
% Invalid input detected at '^' marker.
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/0
Router(config-if)#description LINK2-T0-HUAWEI
Router(config-if)#ip address 10.10.11.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 6 04:05:55.712: %LINK-3-UPDOWN: Interface GigabitEthernet0/0,
changed state to up
*Jun 6 04:05:56.713: %LINEPROTO-5-UPDOWN: Line protocol on Interfa
ce GigabitEthernet0/0, changed state to up
```

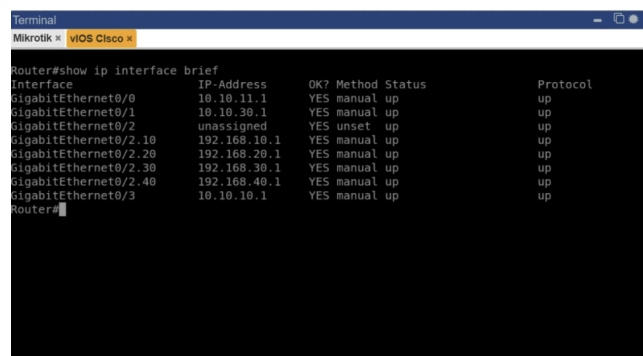
Gambar 20: Konfigurasi Gi0/1 Cisco Router



```
admin@MikroTik> ip address add address=10.10.30.2/30 interface=ether2 comment=T0
admin@MikroTik> ip address add address=10.10.30.2/30 interface=ether2 comment=T
admin@MikroTik> ip address add address=10.10.30.2/30 interface=ether2 comment=
admin@MikroTik> ip address add address=10.10.30.2/30 interface=ether2 comment=t
admin@MikroTik> ip address add address=10.10.30.2/30 interface=ether2 comment=
admin@MikroTik> ip address add address=10.10.30.2/30 interface=ether2 commen
admin@MikroTik> ip address add address=10.10.30.2/30 interface=ether2 comm
admin@MikroTik> ip address add address=10.10.30.2/30 interface=ether2
admin@MikroTik> ip address add address=10.10.30.2/30 interface=ether2 c
admin@MikroTik> ip address add address=10.10.30.2/30 interface=ether2
admin@MikroTik> ip address add address=10.10.30.2/30 interface=ether2
admin@MikroTik> ip address add address=10.10.30.2/30 interface=ether
admin@MikroTik> ip address add address=10.10.30.2/30 interface=ethe
admin@MikroTik> ip address add address=10.10.30.2/30 interface=eth
admin@MikroTik> ip address add address=10.10.30.2/30 interface=eh
admin@MikroTik> ip address add address=10.10.30.2/30 interface=ehe
admin@MikroTik> ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ::: T0-HUAWEI
10.10.20.2/30 10.10.20.0 ether1
1 ::: T0-CISCO
10.10.30.2/30 10.10.30.0 ether2
admin@MikroTik>
```

Gambar 21: Konfigurasi IP address pada MikroTik

7. Verifikasi *interface* pada Cisco Router, Huawei Router, dan MikroTik



```
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.10.11.1 YES manual up up
GigabitEthernet0/1 10.10.30.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset up up
GigabitEthernet0/2.10 192.168.10.1 YES manual up up
GigabitEthernet0/2.20 192.168.20.1 YES manual up up
GigabitEthernet0/2.30 192.168.30.1 YES manual up up
GigabitEthernet0/2.40 192.168.40.1 YES manual up up
GigabitEthernet0/3 10.10.10.1 YES manual up up
Router#
```

Gambar 22: Verifikasi pada Cisco Router


```

Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.10.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.11.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms

```

Gambar 26: Huawei Router ping ke Cisco Link 1 dan Link 2

```

Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.11.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
  Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=2 ms
  Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

--- 10.10.20.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]

```

Gambar 27: Huawei Router ping ke MikroTik

```

Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
[admin@MikroTik] > ping 10.10.20.1
  SEQ HOST                                SIZE TTL TIME  ST..
  0 10.10.20.1                            56 255 3ms
  1 10.10.20.1                            56 255 1ms
  2 10.10.20.1                            56 255 1ms
  3 10.10.20.1                            56 255 2ms
  4 10.10.20.1                            56 255 1ms
  5 10.10.20.1                            56 255 1ms
  6 10.10.20.1                            56 255 1ms
  7 10.10.20.1                            56 255 5ms
  8 10.10.20.1                            56 255 1ms
  9 10.10.20.1                            56 255 1ms
 10 10.10.20.1                            56 255 1ms
sent=11 received=11 packet-loss=0% min-rtt=1ms avg-rtt=1ms
max-rtt=5ms

[admin@MikroTik] > ping 10.10.30.1
  SEQ HOST                                SIZE TTL TIME  ST..
  0 10.10.30.1                            56 255 1ms
  1 10.10.30.1                            56 255 1ms
  2 10.10.30.1                            56 255 1ms
  3 10.10.30.1                            56 255 1ms
  4 10.10.30.1                            56 255 1ms
  5 10.10.30.1                            56 255 1ms
sent=6 received=6 packet-loss=0% min-rtt=1ms avg-rtt=1ms
max-rtt=1ms

[admin@MikroTik] >

```

Gambar 28: MikroTik ping ke Huawei dan Cisco

1.4 Konfigurasi OSPF Multivendor dan OSPF Cost

1. Masuk ke Cisco Router, lalu konfigurasi OSPF dengan *router-id* 1.1.1.1 dan iklankan seluruh *network* terkait

```

Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
.....
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/8 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.....
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/6 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.....
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/2 ms
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 10.10.10.0
% Incomplete command.

Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#network 10.10.11.0 0.0.0.3 area 0
Router(config-router)#network 10.10.30.0 0.0.0.3 area 0
Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
Router(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router(config-router)#network 192.168.40.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#

```

Gambar 29: Konfigurasi OSPF Cisco Router

2. Atur *OSPF cost* pada *interface* Cisco, yaitu *cost* 10 untuk *link* menuju Huawei dan *cost* 100 untuk *link* menuju MikroTik

```

Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
Router(config)#interface gi0/0
Router(config-if)#description LINK2-T0-HUAWEI
Router(config-if)#ip ospf cost 10
Router(config-if)#exit
Router(config)#interface gi0/1
Router(config-if)#description LINK-T0-MIKROTIK
Router(config-if)#ip ospf cost 100
Router(config-if)#exit
Router(config)#exit
Router#show
*Jun 6 04:27:14.789: %SYS-5-CONFIG_I: Configured from console by console ip
% Incomplete command.

Router#show ip ospf neighbor
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR
Gateway of last resort is not set

```

Gambar 30: Konfigurasi OSPF cost pada interface Cisco dan verifikasi

```

Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
Router#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 110)

```

Gambar 31: Verifikasi route pada Cisco

3. Masuk ke Huawei Router, lalu konfigurasi OSPF dengan *router-id* 2.2.2.2 dan iklankan seluruh *network* terkait
4. Atur *OSPF cost* pada *interface* Huawei dengan pola yang sama seperti Cisco Router

```

Terminal
Mikrotik * vIOS Cisco * NE40E Huawei *
[*HUAWEI]interface Ethernet1/0/3
[*HUAWEI-Ethernet1/0/3]description LINK2-TO-MIKROTIK
[*HUAWEI-Ethernet1/0/3]ospf cost 10
[*HUAWEI-Ethernet1/0/3]ospf cost 100
[*HUAWEI-Ethernet1/0/3]quit
[*HUAWEI]commit
[*HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2
Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: 10.10.10.2 MTU: 1500
Dead timer due in 34 sec
Retrans timer interval: 5
Neighbor is up for 00h00m11s
Neighbor Up Time : 2026-06-06 04:33:45
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 36 sec
Retrans timer interval: 5
Neighbor is up for 00h00m11s
Neighbor Up Time : 2026-06-06 04:33:45
Authentication Sequence: [ 0 ]
[~HUAWEI]

```

Gambar 32: Konfigurasi OSPF cost pada Huawei dan verifikasi

```

Terminal
Mikrotik * vIOS Cisco * NE40E Huawei *
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2
[~HUAWEI]display ospf routing

OSPF Process 1 with Router ID 2.2.2.2
Routing Tables

Routing for Network
Destination Cost Type NextHop AdvRouter Area
10.10.10.0/30 10 Direct 10.10.10.2 2.2.2.2 0.0.0.0
10.10.11.0/30 10 Direct 10.10.11.2 2.2.2.2 0.0.0.0
10.10.20.0/30 100 Direct 10.10.20.1 2.2.2.2 0.0.0.0
10.10.30.0/30 110 Stub 10.10.11.1 1.1.1.1 0.0.0.0
10.10.30.0/30 110 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.50.0/24 1 Direct 192.168.50.1 2.2.2.2 0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0

OSPF Process 65534 with Router ID 128.1.138.137
Routing Tables

Routing for Network
Destination Cost Type NextHop AdvRouter Area
128.1.138.137/32 0 Direct 128.1.138.137 128.1.138.137 0.0.0.0

Total Nets: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0

```

Gambar 33: Verifikasi route pada Huawei

5. Masuk ke MikroTik, lalu konfigurasi *router-id* OSPF menjadi 3.3.3.3
6. Tambahkan *network* OSPF pada MikroTik dan atur *cost interface* sebagai jalur *backup*


```

Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x

Router con0 is now available

Press RETURN to get started.

*****
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* education. IOSv is provided as-is and is not supported by Cisco's *
* Technical Advisory Center. Any use or disclosure, in whole or in part, *
* of the IOSv Software or Documentation to any third party for any *
* purposes is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****

Router>enable
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/BDR 00:00:31 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/BDR 00:00:39 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/BDR 00:00:30 10.10.10.2 GigabitEthernet0/3
Router#

```

Gambar 37: Verifikasi OSPF neighbor pada Cisco

```

Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x

[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
State: Full            Mode:Nbr is Slave    Priority: 1
DR: 10.10.10.1        BDR: 10.10.10.2      MTU: 1500
Dead timer due in 36 sec
Retrans timer interval: 5
Neighbor is up for 00h06m55s
Neighbor Up Time : 2026-06-06 04:33:45
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3      Address: 10.10.20.2
State: Full            Mode:Nbr is Master   Priority: 1
DR: 10.10.20.1        BDR: 10.10.20.2      MTU: 1500
Dead timer due in 39 sec
Retrans timer interval: 5
Neighbor is up for 00h02m19s
Neighbor Up Time : 2026-06-06 04:38:21
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
State: Full            Mode:Nbr is Slave    Priority: 1
DR: 10.10.11.1        BDR: 10.10.11.2      MTU: 1500
Dead timer due in 34 sec
Retrans timer interval: 5
Neighbor is up for 00h06m55s

```

Gambar 38: Verifikasi OSPF neighbor pada Huawei

```

Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
State: Full            Mode:Nbr is Slave    Priority: 1
DR: 10.10.10.1        BDR: 10.10.10.2      MTU: 1500
Dead timer due in 36 sec
Retrans timer interval: 5
Neighbor is up for 00h06m55s
Neighbor Up Time : 2026-06-06 04:33:45
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3      Address: 10.10.20.2
State: Full            Mode:Nbr is Master   Priority: 1
DR: 10.10.20.1        BDR: 10.10.20.2      MTU: 1500
Dead timer due in 39 sec
Retrans timer interval: 5
Neighbor is up for 00h02m19s
Neighbor Up Time : 2026-06-06 04:38:21
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
State: Full            Mode:Nbr is Slave    Priority: 1
DR: 10.10.11.1        BDR: 10.10.11.2      MTU: 1500
Dead timer due in 34 sec
Retrans timer interval: 5
Neighbor is up for 00h06m55s
Neighbor Up Time : 2026-06-06 04:33:45
Authentication Sequence: [ 0 ]

```

Gambar 39: Detail OSPF neighbor Huawei

```

Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x
0 instance=default router-id=1.1.1.1 address=10.10.30.1
  interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=23s
1 instance=default router-id=2.2.2.2 address=10.10.20.1
  interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=10.10.20.2 state="Full" state-changes=5
[admin@Mikrotik] > routing ospf interface add interface=ether2 cost=100
failure: there already exists configuration for this interface cost=100
[admin@Mikrotik] > ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY
0 ADo 10.10.10.0/30 10.10.20.1
1 ADo 10.10.11.0/30 10.10.30.1
2 ADo 192.168.10.0/24 10.10.30.1
3 ADo 192.168.20.0/24 10.10.30.1
4 ADo 192.168.30.0/24 10.10.30.1
5 ADo 192.168.40.0/24 10.10.30.1
6 ADo 192.168.50.0/24 10.10.20.1
[admin@Mikrotik] > routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1
  interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=3m25s
1 instance=default router-id=2.2.2.2 address=10.10.20.1
  interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=3m25s
[admin@Mikrotik] >

```

Gambar 40: Verifikasi OSPF neighbor pada MikroTik

8. Verifikasi tabel *routing* OSPF pada ketiga perangkat

```

Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x
*****
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* education. IOSv is provided as is and is not supported by Cisco's *
* Technical Advisory Center. Any use or disclosure, in whole or in part, *
* of the IOSv Software or Documentation to any third party for any *
* purposes is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****
Router>enable
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/BDR 00:00:31 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/BDR 00:00:39 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/BDR 00:00:30 10.10.10.2 GigabitEthernet0/3
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:08:22 ago
Routing Descriptor Blocks:
  10.10.11.2, from 2.2.2.2, 00:08:22 ago, via GigabitEthernet0/0
    Route metric is 11, traffic share count is 1
  * 10.10.10.2, from 2.2.2.2, 00:08:22 ago, via GigabitEthernet0/3
    Route metric is 11, traffic share count is 1

```

Gambar 41: Cek route menuju Huawei pada Cisco

```

Router>enable
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/BDR 00:00:31 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/BDR 00:00:39 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/BDR 00:00:30 10.10.10.2 GigabitEthernet0/3
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:08:22 ago
Routing Descriptor Blocks:
  10.10.11.2, from 2.2.2.2, 00:08:22 ago, via GigabitEthernet0/0
    Route metric is 11, traffic share count is 1
  * 10.10.10.2, from 2.2.2.2, 00:08:22 ago, via GigabitEthernet0/3
    Route metric is 11, traffic share count is 1
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O 10.10.20.0/30 [110/110] via 10.10.11.2, 00:08:51, GigabitEthernet0/0
O 110/110 via 10.10.10.2, 00:08:51, GigabitEthernet0/3
O 192.168.50.0/24 [110/11] via 10.10.11.2, 00:08:51, GigabitEthernet0/0
O 110/11 via 10.10.10.2, 00:08:51, GigabitEthernet0/3
Router#

```

Gambar 42: Verifikasi route OSPF pada Cisco

```

terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
Error: Unrecognized command found at '^' position.
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask  Proto  Pre  Cost    Flags NextHop        Interface
-----
10.10.30.0/30    OSPF   10   110     D   10.10.11.1  Ethernet1/0/4
10.10.30.0/30    OSPF   10   110     D   10.10.10.1  Ethernet1/0/0
192.168.10.0/24  OSPF   10   11       D   10.10.11.1  Ethernet1/0/4
192.168.10.0/24  OSPF   10   11       D   10.10.10.1  Ethernet1/0/0
192.168.20.0/24  OSPF   10   11       D   10.10.11.1  Ethernet1/0/4
192.168.20.0/24  OSPF   10   11       D   10.10.10.1  Ethernet1/0/0
192.168.30.0/24  OSPF   10   11       D   10.10.11.1  Ethernet1/0/4
192.168.30.0/24  OSPF   10   11       D   10.10.10.1  Ethernet1/0/0
192.168.40.0/24  OSPF   10   11       D   10.10.11.1  Ethernet1/0/4
192.168.40.0/24  OSPF   10   11       D   10.10.10.1  Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4
---- More ----

```

Gambar 43: Verifikasi route OSPF pada Huawei (1)

```

terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
[~HUAWEI]
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask  Proto  Pre  Cost    Flags NextHop        Interface
-----
10.10.30.0/30    OSPF   10   110     D   10.10.11.1  Ethernet1/0/4
10.10.30.0/30    OSPF   10   110     D   10.10.10.1  Ethernet1/0/0
192.168.10.0/24  OSPF   10   11       D   10.10.11.1  Ethernet1/0/4
192.168.10.0/24  OSPF   10   11       D   10.10.10.1  Ethernet1/0/0
192.168.20.0/24  OSPF   10   11       D   10.10.11.1  Ethernet1/0/4
192.168.20.0/24  OSPF   10   11       D   10.10.10.1  Ethernet1/0/0
192.168.30.0/24  OSPF   10   11       D   10.10.11.1  Ethernet1/0/4
192.168.30.0/24  OSPF   10   11       D   10.10.10.1  Ethernet1/0/0
192.168.40.0/24  OSPF   10   11       D   10.10.11.1  Ethernet1/0/4
192.168.40.0/24  OSPF   10   11       D   10.10.10.1  Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask  Proto  Pre  Cost    Flags NextHop        Interface
-----
10.10.10.0/30    OSPF   10   10       10.10.10.2  Ethernet1/0/0
10.10.11.0/30    OSPF   10   10       10.10.11.2  Ethernet1/0/4
10.10.20.0/30    OSPF   10  100       10.10.20.1  Ethernet1/0/3
192.168.50.0/24  OSPF   10   1         192.168.50.1 Ethernet1/0/2

[~HUAWEI]
[~HUAWEI]
[~HUAWEI]

```

Gambar 44: Verifikasi route OSPF pada Huawei (2)

```

terminal
Mikrotik x vIOS Cisco x NE40E Huawei x
# DST-ADDRESS      PREF-SRC  GATEWAY
0 ADo 10.10.10.0/30
1 ADo 10.10.11.0/30
2 ADo 192.168.10.0/24
3 ADo 192.168.20.0/24
4 ADo 192.168.30.0/24
5 ADo 192.168.40.0/24
6 ADo 192.168.50.0/24

[admin@MikroTik] > routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1
  interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=3m25s
1 instance=default router-id=2.2.2.2 address=10.10.20.1
  interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=3m25s

[admin@MikroTik] > ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC  GATEWAY
0 ADo 10.10.10.0/30
1 ADo 10.10.11.0/30
2 ADo 192.168.10.0/24
3 ADo 192.168.20.0/24
4 ADo 192.168.30.0/24
5 ADo 192.168.40.0/24
6 ADo 192.168.50.0/24

[admin@MikroTik] >

```

Gambar 45: Verifikasi route OSPF pada MikroTik

9. Lakukan pengujian konektivitas *end-to-end* dari setiap PC VLAN menuju LAN Huawei, dan sebaliknya

```
Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x VPC PC-FIN... x
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=10.585 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.004 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=7.077 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=5.717 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=7.570 ms
VPCS>
```

Gambar 46: PC VLAN 10 (Finance) ping ke PC LAN Huawei

```
Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=6.908 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.662 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=9.470 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.718 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=7.124 ms
VPCS>
```

Gambar 47: PC VLAN 20 (HR) ping ke PC LAN Huawei

```
Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x VPC PC-ITV... x
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.035 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.530 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.647 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.097 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.885 ms
VPCS>
```

Gambar 48: PC VLAN 30 (IT) ping ke PC LAN Huawei

```
Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x VPC PC-ITV... x VPC PC-MA... x
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.845 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.951 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=7.017 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=8.036 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=6.069 ms
VPCS>
```

Gambar 49: PC VLAN 40 (Marketing) ping ke PC LAN Huawei

```
Terminal
Mikrotik x vIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x VPC PC-ITV... x VPC PC-MA... x VPC x
VPCS> ping 192.168.10.11
84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=7.750 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=7.276 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=6.034 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=7.772 ms
84 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=18.920 ms
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=7.928 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=11.389 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=8.516 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=6.475 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=62 time=5.919 ms
VPCS> ping 192.168.30.11
192.168.30.11 icmp_seq=1 timeout
192.168.30.11 icmp_seq=2 timeout
192.168.30.11 icmp_seq=3 timeout
192.168.30.11 icmp_seq=4 timeout
192.168.30.11 icmp_seq=5 timeout
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=5.812 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=9.049 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=7.474 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=6.173 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=9.131 ms
VPCS>
```

Gambar 50: PC LAN Huawei ping ke VLAN 10, 20, 30, dan 40

1.5 Pengujian Failover OSPF

1. Pada kondisi normal dengan dua *link* Cisco-Huawei aktif, cek *route* dari Cisco menuju LAN Huawei dan lakukan *trace* dari PC VLAN menuju LAN Huawei

```
VPCS>
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    4.972 ms  3.289 ms  3.573 ms
 2  10.10.11.2     8.839 ms  4.800 ms  5.527 ms
 3  *192.168.50.10 11.816 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>
```

Gambar 51: Trace dari PC VLAN 10 ke LAN Huawei

2. Matikan salah satu *link* Cisco-Huawei, lalu cek *route* dan lakukan *trace* kembali

```
Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x VPC PC-IT-V... x VPC PC-MA... x VPC x
0  192.168.50.0/24 [110/11] via 10.10.10.2, 00:08:51, GigabitEthernet0/3
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:15:08 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:15:08 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:15:08 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)#shutdown
Router(config-if)#ex1
*Jun 6 04:50:15.186: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from FULL to DOWN, Neighbor Down: Interface down or detached
Router(config)#
*Jun 6 04:50:17.154: %LINK-5-CHANGED: Interface GigabitEthernet0/3, changed state to administratively down
*Jun 6 04:50:18.154: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to down
Router(config)#exit
Router#
*Jun 6 04:50:23.296: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:17:15 ago
  Routing Descriptor Blocks:
    * 10.10.11.2, from 2.2.2.2, 00:17:15 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
Router#
```

Gambar 52: Shutdown Gi0/3 dan cek route pada Cisco

```
3  192.168.50.10 11.816 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    3.751 ms  4.534 ms  5.273 ms
 2  10.10.11.2     7.071 ms  3.725 ms  4.792 ms
 3  *192.168.50.10 8.447 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>
```

Gambar 53: Trace dari PC VLAN ke LAN Huawei

3. Matikan *link* Cisco-Huawei kedua, lalu cek *route* dan lakukan *trace* untuk memastikan jalur berpindah ke MikroTik sebagai *backup*

```
Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x VPC PC-ITV... x VPC PC-MA... x VPC x
Router(config)#exit
Router#
*Jun 6 04:50:23.296: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:17:15 ago
  Routing Descriptor Blocks:
    * 10.10.11.2, from 2.2.2.2, 00:17:15 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/0
Router(config-if)#shutdown
Router(config-if)#
*Jun 6 04:51:42.084: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
*Jun 6 04:51:44.053: %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down
*Jun 6 04:51:45.053: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down
Router(config-if)#exit
Router(config)#exit
Router#show i
*Jun 6 04:51:53.608: %SYS-5-CONFIG_I: Configured from console by console
% Incomplete command.
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 10.10.30.2 on GigabitEthernet0/1, 00:00:18 ago
  Routing Descriptor Blocks:
    * 10.10.30.2, from 2.2.2.2, 00:00:18 ago, via GigabitEthernet0/1
      Route metric is 201, traffic share count is 1
Router#
```

Gambar 54: Shutdown Gi0/0 dan cek route pada Cisco

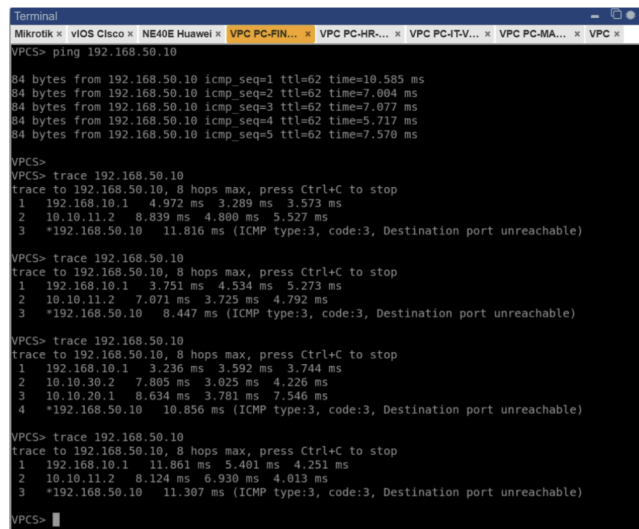
```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    3.236 ms  3.592 ms  3.744 ms
 2  10.10.30.2     7.805 ms  3.025 ms  4.226 ms
 3  10.10.20.1     8.634 ms  3.781 ms  7.546 ms
 4  *192.168.50.10 10.856 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>
```

Gambar 55: Trace dari PC VLAN ke LAN Huawei

4. Aktifkan kembali kedua *link* Cisco-Huawei, lalu cek *route* dan lakukan *trace* untuk memastikan jalur kembali ke *link* utama

```
Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x VPC PC-ITV... x VPC PC-MA... x VPC x
Router(config)#interface gi0/0
*Jun 6 04:52:58.468: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
Router(config-if)#interface gi0/3
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 6 04:53:12.395: %LINK-3-UPDOWN: Interface GigabitEthernet0/3, changed state to up
*Jun 6 04:53:13.395: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to up
% Ambiguous command: "e"
Router(config)#
Router(config)#
Router(config)#exit
Router#
*Jun 6 04:53:17.892: %SYS-5-CONFIG_I: Configured from console by console
% Type "show ?" for a list of subcommands
Router#
*Jun 6 04:53:22.320: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOADING to FULL, Loading Done
% Type "show ?" for a list of subcommands
Router#
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/3, 00:00:10 ago
  Routing Descriptor Blocks:
    * 10.10.11.2, from 2.2.2.2, 00:00:10 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
    * 10.10.11.2, from 2.2.2.2, 00:00:10 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#
```

Gambar 56: Aktifkan Gi0/3 dan Gi0/0 dan cek route pada Cisco



```
Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x VPC PC-ITV... x VPC PC-MA... x VPC x
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=10.585 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.004 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=7.077 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=5.717 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=7.570 ms

VPCS>
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 4.972 ms 3.289 ms 3.573 ms
 2 10.10.11.2 8.839 ms 4.800 ms 5.527 ms
 3 *192.168.50.10 11.816 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.751 ms 4.534 ms 5.273 ms
 2 10.10.11.2 7.071 ms 3.725 ms 4.792 ms
 3 *192.168.50.10 8.447 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.236 ms 3.592 ms 3.744 ms
 2 10.10.20.2 7.805 ms 3.025 ms 4.226 ms
 3 10.10.20.1 8.634 ms 3.781 ms 7.546 ms
 4 *192.168.50.10 10.856 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 11.861 ms 5.481 ms 4.251 ms
 2 10.10.11.2 8.124 ms 6.930 ms 4.013 ms
 3 *192.168.50.10 11.307 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> 
```

Gambar 57: Trace dari PC VLAN ke LAN Huawei

2 Analisis Hasil Percobaan

2.1 Praktikum 1 – Konfigurasi Cisco Switch

Pada praktikum ini, Cisco Switch berhasil dikonfigurasi dengan empat VLAN, yaitu VLAN 10 (Finance), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (Marketing). Setiap VLAN diberikan access port yang terhubung ke masing-masing PC end-device, sehingga traffic antar VLAN terisolasi secara logis meskipun berada pada switch fisik yang sama. Port menuju Cisco Router dikonfigurasi sebagai trunk port menggunakan enkapsulasi IEEE 802.1Q, yang memungkinkan keempat VLAN dibawa sekaligus melalui satu koneksi fisik. Hasil verifikasi menggunakan perintah `show vlan brief` dan `show interfaces trunk` menunjukkan bahwa seluruh VLAN aktif dan trunk port telah membawa VLAN 10, 20, 30, dan 40 sesuai konfigurasi.

2.2 Praktikum 2 – Konfigurasi Cisco Router

Cisco Router dikonfigurasi menggunakan metode Router-on-a-Stick (ROAS), yaitu satu interface fisik Gi0/2 dibagi menjadi empat subinterface yang masing-masing menjadi gateway untuk satu VLAN. Subinterface Gi0/2.10, Gi0/2.20, Gi0/2.30, dan Gi0/2.40 dikonfigurasi dengan enkapsulasi dot1Q sesuai VLAN ID masing-masing. VLAN 10 dan VLAN 20 dikonfigurasi menggunakan DHCP server pada router, sehingga PC pada kedua VLAN tersebut mendapatkan IP address secara otomatis. Sementara itu, PC pada VLAN 30 dan VLAN 40 menggunakan IP static sesuai tabel addressing. Hasil pengujian menunjukkan bahwa seluruh PC berhasil melakukan ping ke gateway masing-masing, dan ping antar-VLAN pun berhasil dilakukan, membuktikan bahwa inter-VLAN routing melalui Cisco Router berfungsi dengan baik.

2.3 Praktikum 3 – Konfigurasi Huawei Router dan IP Address Antarrouter

Pada praktikum ini, seluruh link antarrouter berhasil dikonfigurasi dengan IP address sesuai tabel addressing. LAN pada sisi Huawei dengan network 192.168.50.0/24 berhasil diaktifkan melalui interface Ethernet1/0/2, dan PC LAN Huawei berhasil melakukan ping ke gateway 192.168.50.1.

Dua link fisik antara Cisco dan Huawei berhasil dikonfigurasi, yaitu link pertama melalui Gi0/3 – Ethernet1/0/0 pada network 10.10.10.0/30, dan link kedua melalui Gi0/0 – Ethernet1/0/4 pada network 10.10.11.0/30. Link antara Huawei dan MikroTik dikonfigurasi pada network 10.10.20.0/30, serta link antara Cisco dan MikroTik pada network 10.10.30.0/30. Verifikasi ping antarrouter menunjukkan bahwa seluruh router dapat saling berkomunikasi dengan router tetangganya secara langsung.

2.4 Praktikum 4 – Konfigurasi OSPF Multivendor dan OSPF Cost

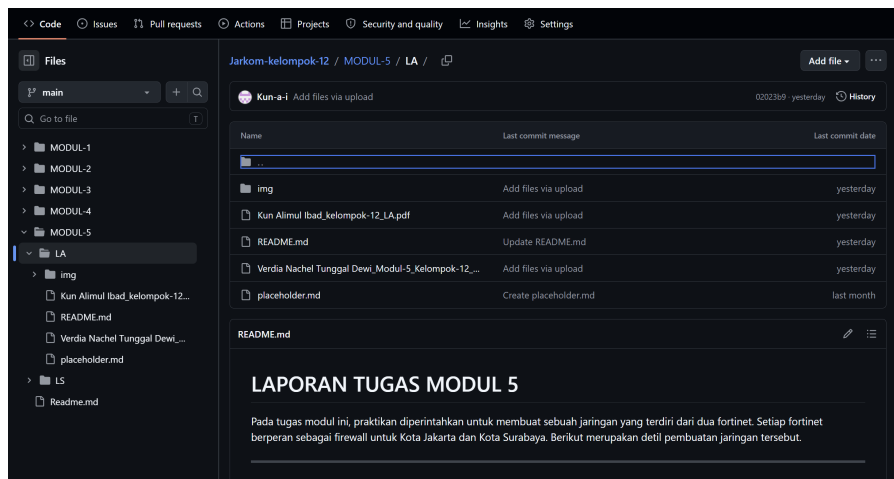
OSPF berhasil dikonfigurasi pada tiga perangkat dari vendor berbeda, yaitu Cisco, Huawei, dan MikroTik, semuanya dalam area 0 (backbone). Cisco Router menggunakan router-id 1.1.1.1, Huawei menggunakan 2.2.2.2, dan MikroTik menggunakan 3.3.3.3. Dua link utama Cisco–Huawei dikonfigurasi dengan OSPF cost 10, sedangkan link melalui MikroTik dikonfigurasi dengan cost 100, sehingga MikroTik berfungsi sebagai jalur backup. Hasil verifikasi menunjukkan bahwa Cisco Router berhasil membentuk neighbor dengan Huawei melalui dua link sekaligus, serta dengan MikroTik melalui satu link. Tabel routing OSPF pada masing-masing router menunjukkan bahwa seluruh network telah terdistribusi dengan benar. Pengujian end-to-end dari PC VLAN 10, 20, 30, dan 40 menuju PC LAN Huawei 192.168.50.10 berhasil dilakukan, begitu pula sebaliknya.

2.5 Praktikum 5 – Pengujian Failover OSPF

Pengujian failover dilakukan dalam tiga skenario. Pada kondisi normal dengan dua link Cisco–Huawei aktif, hasil trace menunjukkan bahwa traffic melewati jalur langsung antara Cisco dan Huawei dengan cost terendah. Pada kondisi satu link dimatikan (Gi0/3 shutdown), OSPF secara otomatis melakukan konvergensi dan traffic tetap melewati jalur langsung Cisco–Huawei melalui link yang masih aktif, tanpa berpindah ke jalur backup MikroTik. Hal ini sesuai dengan desain karena kedua link utama memiliki cost yang sama, sehingga satu link masih cukup untuk mempertahankan jalur optimal. Pada kondisi kedua link dimatikan sekaligus, OSPF melakukan reconverge dan traffic berpindah ke jalur backup melalui MikroTik dengan jalur PC VLAN → Cisco Router → MikroTik → Huawei Router → LAN Huawei. Setelah kedua link diaktifkan kembali, routing kembali berpindah ke jalur utama Cisco–Huawei, membuktikan bahwa mekanisme failover dan failback OSPF berjalan dengan baik.

3 Hasil Tugas Modul

ada di readme.md github



Gambar 58: Tugas Modul

4 Kesimpulan

Berdasarkan hasil seluruh praktikum yang telah dilakukan, dapat disimpulkan beberapa hal sebagai berikut.

Pertama, teknologi VLAN berhasil diimplementasikan pada Cisco Switch untuk membagi jaringan fisik menjadi empat segmen logis yang terisolasi. Konfigurasi access port dan trunk port berjalan sesuai standar IEEE 802.1Q, sehingga traffic antar-VLAN tidak bercampur dan keamanan serta efisiensi jaringan dapat terjaga.

Kedua, metode Router-on-a-Stick (ROAS) pada Cisco Router terbukti efektif sebagai solusi inter-VLAN routing menggunakan satu interface fisik dengan beberapa subinterface. Fitur DHCP server yang dikonfigurasi pada router juga berjalan dengan baik, terbukti dari PC VLAN 10 dan VLAN 20 yang berhasil mendapatkan IP address secara otomatis dan dapat berkomunikasi lintas VLAN.

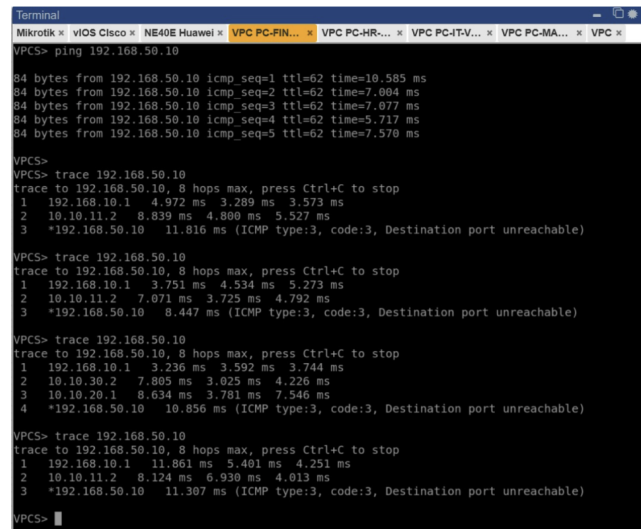
Ketiga, konfigurasi IP address pada seluruh link antarrouter, baik Cisco–Huawei, Huawei–MikroTik, maupun Cisco–MikroTik, berhasil dilakukan dan seluruh router dapat saling berkomunikasi dengan router tetangganya secara langsung. Hal ini menjadi fondasi yang diperlukan sebelum protokol routing dinamis dapat dijalankan.

Keempat, OSPF berhasil dikonfigurasi secara multivendor pada tiga perangkat dari vendor yang berbeda, yaitu Cisco, Huawei, dan MikroTik. Meskipun setiap vendor memiliki sintaks konfigurasi yang berbeda, OSPF tetap dapat membentuk neighbor relationship dan mendistribusikan informasi routing ke seluruh jaringan karena menggunakan standar protokol yang sama. Pengaturan OSPF cost juga terbukti berfungsi dalam menentukan jalur utama dan jalur backup sesuai desain topologi.

Kelima, mekanisme failover OSPF berjalan sesuai yang diharapkan. Ketika satu link utama Cisco–Huawei dimatikan, traffic tetap menggunakan link utama yang tersisa tanpa berpindah ke jalur backup. Ketika kedua link utama dimatikan, OSPF secara otomatis melakukan reconverge dan memindahkan traffic ke jalur backup melalui MikroTik. Setelah link utama diaktifkan kembali, routing kembali berpindah ke jalur optimal semula. Perilaku ini membuktikan bahwa OSPF mampu menjamin ketersediaan jaringan (*high availability*) secara dinamis tanpa intervensi manual.

5 Lampiran

5.1 Dokumentasi saat praktikum



```
Terminal
Mikrotik x VIOS Cisco x NE40E Huawei x VPC PC-FIN... x VPC PC-HR... x VPC PC-ITV... x VPC PC-MA... x VPC x
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=10.585 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.004 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=7.077 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=5.717 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=7.570 ms

VPCS>
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 4.972 ms 3.289 ms 3.573 ms
 2 10.10.11.2 8.839 ms 4.800 ms 5.527 ms
 3 *192.168.50.10 11.816 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.751 ms 4.534 ms 5.273 ms
 2 10.10.11.2 7.071 ms 3.725 ms 4.792 ms
 3 *192.168.50.10 8.447 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.236 ms 3.592 ms 3.744 ms
 2 10.10.30.2 7.805 ms 3.025 ms 4.226 ms
 3 10.10.20.1 8.634 ms 3.781 ms 7.546 ms
 4 *192.168.50.10 10.856 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 11.861 ms 5.401 ms 4.251 ms
 2 10.10.11.2 8.124 ms 6.030 ms 4.013 ms
 3 *192.168.50.10 11.307 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> |
```

Gambar 59: Gambar Saat Praktikum, Maaf Mas/Mba Gaada Foto Lain